

A sharp cover bound for congruences of finite lattices

Pairwise joins, equality, and bounded skeletons

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Abstract

Let L be a finite lattice with n elements. We give an elementary proof that, if one element has at least k upper covers and $k \geq 3$, then $|\text{Con}(L)| \leq 2^{n-k-1}$. This supplies the strengthening suggested immediately after Lemma 4.4 of Czédli's *Accumulation points of congruence densities of finite lattices*, replacing its multicolor Ramsey threshold by k itself. The proof counts two complementary losses: join-reducible elements and repeated principal-congruence labels. Equalities between pairwise joins control both losses. We also characterize equality: precisely a length-two lattice with k atoms, with an arbitrary chain attached at either end. Among nonextremal lattices with such a cover configuration, the sharp next bound is $3 \cdot 2^{n-k-3}$. Finally, for $\text{cd}(L) \geq p > 0$ and $t = \lfloor \log_2(1/p) \rfloor$, the skeleton introduced by Czédli has at most $2t(\max\{2, t\} + 1)$ elements. All necessary congruence tools are proved here. Exact exhaustive tests cover every isomorphism class through eight elements, represented by 4,008 naturally labelled lattices, and check 6,292 cover configurations. The archive contains the programs and their recorded output.

Status. This is a proposed research solution with complete written proofs, not an independently refereed or proof-assistant-verified result. The literature search found the stated question unresolved in the consulted sources; it does not certify priority. The elementary background, the source's existing special cases, and the contributions of this draft are distinguished below.

Keywords. Finite lattice; lattice congruence; upper cover; join-irreducible element; congruence density; extremal lattice; skeleton.

Selection record. A single operating-system random draw from 80 mathematical areas selected number 10, *lattice theory*. No redraw was made. The complete list and the exclusion audit are included in the package.

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1 The question and the results

1.1 A precise published target

A *lattice* is a partially ordered set in which any two elements x, y have a greatest lower bound $x \wedge y$ and a least upper bound $x \vee y$. An element a *covers* u , written $u \prec a$, when $u < a$ and no element lies strictly between them. Throughout, lattices are finite and nonempty. A *congruence* is an equivalence relation compatible with both lattice operations. We write $\text{Con}(L)$ for the set of congruences of L .

The normalization

$$\text{cd}(L) = \frac{|\text{Con}(L)|}{2^{|L|-1}} \quad (1.1)$$

is the congruence density used by Czédli. The denominator is the classical maximum for a lattice of that size, attributed to Freese [2]; a self-contained proof appears in Section 2.

For $k \geq 3$, let $R_{k-1}(k) = R(k, \dots, k)$ be the Ramsey number with $k-1$ colors. Czédli proves that $R_{k-1}(k)$ upper covers of one element force $\text{cd}(L) \leq 2^{-k}$, and immediately suggests that k upper covers might already suffice [1, Lemma 4.4 and the following sentence, p. 8]. The precise target of this article is therefore

$$k \geq 3, \quad |\text{UpCov}_L(u)| \geq k \implies \text{cd}(L) \leq 2^{-k}. \quad (1.2)$$

This is an informal suggestion in the source, not a numbered conjecture. We occasionally call it the *k-cover question* as a local shorthand. In particular, this article is not about covering systems of integers, geometric lattices of Euclidean space, or the much broader congruence-lattice representation problem.

1.2 The sharp theorem

Let C_s denote the chain with s elements, including C_1 . Let M_k consist of a bottom u , a top v , and k pairwise incomparable intermediate elements $a_1, \dots, a_k$. Thus $|M_k| = k + 2$. The *glued sum* $A \dot{+} B$ identifies the top of A with the bottom of B and puts every element of A below every element of B .

Theorem 1.1 (Sharp cover bound and all equality cases). *Let $k \geq 3$, and let L be an n -element lattice with an element having at least k upper covers. Then*

$$|\text{Con}(L)| \leq 2^{n-k-1}. \quad (1.3)$$

Equality holds if and only if

$$L \cong C_s \dot{+} M_k \dot{+} C_t \quad (s, t \geq 1, \quad s + t = n - k). \quad (1.4)$$

The same statements hold with “lower covers” in place of “upper covers”. For every $n \geq k + 2$, the bound is attained.

The proof of the inequality is in Section 3; the equality analysis is in Section 4. The proof does not assume modularity, semimodularity, distributivity of L , or that u is the bottom element. Nor does it require the upper covers to have a common pairwise join. The latter condition is the special case already covered by Czédli’s Lemma 4.3 [1].

There is also a sharp gap. If L is not of the form (1.4), then

$$\text{cd}(L) \leq \frac{3}{4} 2^{-k}. \quad (1.5)$$

For every $n \geq k + 3$, this is attained among n -element lattices with k upper covers. Section 5 proves this assertion; it determines the second largest value in the cover-constrained class, but does not classify all of its witnesses.

1.3 Selection, source status, and attribution

The attached manifest [5] was read in full and treated as an exclusion list. No listed problem concerns the finite-lattice congruence bound considered here. The random draw was performed before selecting this question; the selected area was retained throughout. Appendix A records the method and the 80-area list.

As checked on 20 September 2026, arXiv listed only version 1 of the target paper, dated 12 March 2026. The author's publication list reports its acceptance by *CUBO* on 14 August 2026 and still dates the linked manuscript 12 March 2026 [4]. Searches for the exact title and for the cover suggestion did not locate a subsequent resolution. This is evidence for selecting the problem, not an exhaustive priority determination.

In particular, the $k = 3$ density range overlaps earlier classification work: Czédli's separate paper describes all lattices with density exceeding $3/32$ [3]. No claim is made that every low-parameter consequence below was previously unknown. The contributions proposed here are the uniform all- k proof, its equality argument, the resulting cover-constrained extremal statements, and the quadratic skeleton estimate. The finite congruence theory used to obtain them is classical and is reproved, not claimed as new.

2 Elementary congruence tools

2.1 Irreducible elements and principal congruences

Every finite lattice has a bottom 0 and a top 1. Write $Ji(L)$ for the set of elements having exactly one lower cover. For $j \in Ji(L)$, denote that lower cover by j_* . Write $Jr(L)$ for the elements having at least two lower covers, and $Mr(L)$ for those having at least two upper covers. Thus, with $n = |L|$ and $m = |Jr(L)|$,

$$L = \{0\} \dot{\cup} Ji(L) \dot{\cup} Jr(L), \quad |Ji(L)| = n - 1 - m. \quad (2.1)$$

These are also the familiar algebraic notions. A nonzero element is join-irreducible precisely when it cannot be expressed as a join of two strictly smaller elements. Indeed, a unique lower cover lies above every strictly smaller element. Conversely, two distinct lower covers have join equal to the element they cover. In particular, the join of two incomparable elements belongs to $Jr(L)$; the dual assertion holds for their meet.

For $x, y \in L$, let $\text{con}(x, y)$ be the least congruence identifying x and y . It exists as the intersection of all such congruences. Congruences are ordered by inclusion as binary relations. Their least and greatest elements are the equality relation $\mathbf{0}_L$ and the universal relation $\mathbf{1}_L$.

Lemma 2.1 (Convexity of classes). *Every congruence class in a finite lattice is an interval. In particular, if $x \leq z \leq y$ and $x \equiv y \pmod{\theta}$, then $x \equiv z \equiv y \pmod{\theta}$.*

Proof. Apply meet with z to $x \equiv y$: it gives $x \equiv z$. A congruence class is closed under finite meets and joins, so it has a least and a greatest element. Convexity then identifies it with the interval between them. $\square$

Lemma 2.2 (Transposition without a cover hypothesis). *For all $x, y \in L$,*

$$\text{con}(x \wedge y, x) = \text{con}(y, x \vee y). \quad (2.2)$$

Proof. Joining a pair $x \wedge y \equiv x$ with y gives $y \equiv x \vee y$. Conversely, meeting $y \equiv x \vee y$ with x gives $x \wedge y \equiv x$. Each principal congruence is therefore contained in the other. $\square$

Lemma 2.3 (A join-irreducible representative for every edge). *If $x \prec y$, there is $j \in \text{Ji}(L)$ such that*

$$j \vee x = y, \quad j \wedge x = j_*, \quad \text{con}(x, y) = \text{con}(j_*, j). \quad (2.3)$$

If $a_1, \dots, a_k$ are distinct upper covers of the same u , the representatives $j_1, \dots, j_k$ obtained in this way are distinct.

Proof. Choose j minimal in $\downarrow y \setminus \downarrow x$. Then $j \neq 0$, and every element strictly below j lies below x . If j had two distinct lower covers, their join would be j , forcing $j \leq x$. Thus $j \in \text{Ji}(L)$ and $j_* \leq x$. The inequalities $j \not\leq x$ and $j \leq y$ imply $x < j \vee x \leq y$; the cover hypothesis gives $j \vee x = y$. Also $j \wedge x < j$, so $j \wedge x \leq j_*$, while $j_* \leq j \wedge x$. Equation (2.2) proves the congruence equality. For the final assertion, $j_i = j_h$ would imply $a_i = j_i \vee u = j_h \vee u = a_h$. $\square$

2.2 Signatures and the classical counting bound

Assign to each $j \in \text{Ji}(L)$ the label

$$\gamma_j = \text{con}(j_*, j), \quad Q(L) = \{\gamma_j : j \in \text{Ji}(L)\}, \quad q = |Q(L)|. \quad (2.4)$$

The set $Q(L)$ contains *distinct congruences*: repetitions among the γ_j have been removed. It is a poset under inclusion. For $\theta \in \text{Con}(L)$, define its signature

$$\Sigma(\theta) = \{\gamma \in Q(L) : \gamma \leq \theta\}. \quad (2.5)$$

Lemma 2.4 (Signatures determine congruences). *The map Σ is injective. Consequently*

$$|\text{Con}(L)| \leq 2^q \leq 2^{|\text{Ji}(L)|} = 2^{n-1-m} \leq 2^{n-1}. \quad (2.6)$$

Moreover, every congruence is the join of the principal labels that it contains.

Proof. A congruence is determined by the cover edges it collapses. For comparable $x \leq y$, the relation $x \equiv y$ holds exactly when all edges of a saturated chain from x to y collapse; use Lemma 2.1 for one direction and transitivity for the other. For arbitrary x, y , the relation is equivalent to both $x \equiv x \wedge y$ and $y \equiv x \wedge y$. Lemma 2.3 expresses the principal congruence of each edge as one of the γ_j . Thus its collapse is determined by (2.5). This proves injectivity and the cardinality bounds.

Let ρ be the join of all $\gamma \in \Sigma(\theta)$. Then $\rho \leq \theta$, and every edge collapsed by θ is also collapsed by ρ . The preceding argument gives $\rho = \theta$. $\square$

An s -element chain has exactly 2^{s-1} congruences: choose an arbitrary subset of its $s-1$ adjacent edges to collapse. The resulting blocks are intervals, and interval partitions are compatible with minimum and maximum. This verifies that the denominator in (1.1) really is the maximum.

For the main inequality, injectivity alone is enough. We also give the stronger classical description because it yields a sharp gap and clarifies equality. An *order ideal* of a poset is a downward-closed subset, with the empty set included.

Lemma 2.5 (Prime edge labels and ideal representation). *For an edge $x \prec y$, its principal congruence is join-prime: if $\text{con}(x, y) \leq \theta \vee \phi$, then it is contained in θ or in ϕ . Furthermore,*

$$\Sigma : \text{Con}(L) \longrightarrow \text{Idl}(Q(L)) \quad \text{is a bijection.} \quad (2.7)$$

In particular, $|\text{Con}(L)| = 2^q$ if and only if $Q(L)$ is an antichain.

Proof. The join of two congruences is the equivalence closure of their union. To see compatibility of this closure, apply a lattice operation with one argument fixed to a connecting walk; each step remains in the congruence responsible for that step. Compatibility in both arguments follows by changing one at a time. Thus $x \equiv y \pmod{\theta \vee \phi}$ gives a walk from x to y whose steps belong to θ or to ϕ .

Apply the unary lattice polynomial $z \mapsto (z \vee x) \wedge y$ to the walk. Its image lies in $[x, y] = \{x, y\}$, and its endpoints remain x, y . At least one step changes from one endpoint to the other. That step witnesses $x \equiv y$ in θ or in ϕ . This proves join-primeness.

Every signature is an ideal. Conversely, for an ideal $I \subseteq Q(L)$, put $\theta = \bigvee I$, using $\mathbf{0}_L$ for the empty join. Each $\gamma \in Q(L)$ is an edge label, so finite join-primeness implies that $\gamma \leq \bigvee I$ exactly when $\gamma \leq \delta$ for some $\delta \in I$. Since I is downward closed, this is equivalent to $\gamma \in I$. Therefore $\Sigma(\theta) = I$, proving surjectivity. Injectivity is Lemma 2.4. Finally, all 2^q subsets of a finite poset are ideals exactly when it has no comparability between distinct elements. $\square$

3 Pairwise joins and the cover bound

3.1 A local cover configuration

Fix $u \in L$ and $k \geq 3$ distinct upper covers $a_1, \dots, a_k$ of u . We call the selected configuration a *fan*; the selection need not contain all upper covers of u . Put

$$\alpha_i = \text{con}(u, a_i), \quad r = |\{\alpha_1, \dots, \alpha_k\}|, \quad H = \{a_i \vee a_j : 1 \leq i < j \leq k\}. \quad (3.1)$$

Distinct upper covers are incomparable and have meet u . Therefore

$$a_i \wedge a_j = u, \quad H \subseteq \text{Jr}(L), \quad \alpha_i = \text{con}(a_j, a_i \vee a_j) \quad (i \neq j). \quad (3.2)$$

The last equality is transposition. Notice that $[a_j, a_i \vee a_j]$ need not be a cover interval. That is why the unrestricted form of Lemma 2.2 is useful.

Lemma 3.1 (A third cover below a pairwise join). *For distinct i, j, h , if $a_h \leq a_i \vee a_j$, then*

$$\alpha_h \leq \alpha_i \quad \text{and} \quad \alpha_h \leq \alpha_j. \quad (3.3)$$

Proof. The relation $a_j \equiv a_i \vee a_j \pmod{\alpha_i}$, met with a_h , gives $u \equiv a_h \pmod{\alpha_i}$. Interchange i, j for the other inclusion. $\square$

Lemma 3.2 (Join collisions). *The following statements hold for a fan.*

(i) *If i, j, h are distinct and $a_i \vee a_j = a_i \vee a_h$, then*

$$\alpha_j = \alpha_h \leq \alpha_i. \quad (3.4)$$

(ii) *If i, j, h, ℓ are distinct and $a_i \vee a_j = a_h \vee a_\ell$, then all four labels $\alpha_i, \alpha_j, \alpha_h, \alpha_\ell$ are equal.*

Proof. For (i), write the common join as v . Transposing against a_i gives $\alpha_j = \text{con}(a_i, v) = \alpha_h$. Since $a_h \leq a_i \vee a_j$, Lemma 3.1 gives $\alpha_h \leq \alpha_i$.

For (ii), each of a_h, a_ℓ lies below $a_i \vee a_j$, so its label is contained in both α_i and α_j . Conversely, each of a_i, a_j lies below $a_h \vee a_\ell$, so its label is contained in both α_h and α_ℓ . These inclusions give the assertion. $\square$

The combinatorial point is that distinct congruence labels prevent collisions among the corresponding pairwise joins. This is more informative than treating pairwise joins as unrelated colors in an arbitrary complete-graph coloring.

3.2 The two losses in the exponent

Lemma 3.3 (The join-reducible budget). *For a fan of $k \geq 3$ upper covers with r distinct labels,*

$$m = |\text{Jr}(L)| \geq r. \quad (3.5)$$

More precisely,

$$m \geq \varphi(r), \quad \varphi(r) = \begin{cases} 1, & r = 1, \\ 2, & r = 2, \\ \binom{r}{2}, & r \geq 3. \end{cases} \quad (3.6)$$

Proof. For $r = 1$, the nonempty set H supplies a join-reducible element. For $r = 2$, there are two covers a, b with one label and a cover c with the other: here the hypothesis $k \geq 3$ is essential. If $a \vee b = a \vee c$, Lemma 3.2(i) would equate the two labels. Thus these two joins are distinct members of $\text{Jr}(L)$.

For $r \geq 3$, select one cover from each label class. The $\binom{r}{2}$ pairwise joins of these selected covers are all different. Two incident pairs cannot have the same join by part (i) of the collision lemma; two disjoint pairs cannot have the same join by part (ii). All their joins belong to $\text{Jr}(L)$. Finally, $\binom{r}{2} \geq r$ when $r \geq 3$. $\square$

Lemma 3.4 (The repeated-label budget). *For the same fan,*

$$q \leq |\text{Ji}(L)| - k + r = n - 1 - m - k + r. \quad (3.7)$$

Proof. Lemma 2.3 supplies k distinct elements $j_1, \dots, j_k$ of $\text{Ji}(L)$ whose labels are $\alpha_1, \dots, \alpha_k$. They account for only r distinct labels. Each of the other $|\text{Ji}(L)| - k$ elements can add at most one more. $\square$

Proof of the inequality in Theorem 1.1. Combine the signature bound with the two budgets:

$$\begin{aligned} |\text{Con}(L)| &\leq 2^q \\ &\leq 2^{n-1-m-k+r} \\ &\leq 2^{n-1-k}. \end{aligned}$$

The first loss is the m elements that are not available as nonzero join-irreducible generators. The second loss is the $k - r$ repetitions among the fan's k distinct representatives. Their total is $m + (k - r) \geq k$. A lattice and its order dual have exactly the same congruences, with meet and join interchanged, so the lower-cover statement follows as well. $\square$

Corollary 3.5 (A label-sensitive exponent). *Under the fan hypotheses,*

$$\text{cd}(L) \leq 2^{-m-k+r} \leq 2^{-k-\varphi(r)+r}. \quad (3.8)$$

For example, four distinct fan labels give $\text{cd}(L) \leq 2^{-k-2}$, and five distinct labels give $\text{cd}(L) \leq 2^{-k-5}$.

4 Why equality forces a single diamond

The numerical proof also retains enough information to determine the lattice. This requires distinguishing *equal labels* from *independent labels*: even when the count of labels is optimal, comparabilities between them make some subsets unavailable as signatures.

Lemma 4.1 (Antichain labels strengthen the budget). *Suppose $Q(L)$ is an antichain and the fan has $r \geq 2$ distinct labels. Then $|\text{Jr}(L)| > r$.*

Proof. For $r = 2$, choose distinct covers a, b with label A and a cover c with label $B \neq A$. The joins $a \vee b$, $a \vee c$, and $b \vee c$ are all distinct. The first cannot equal either of the other two, as that would give $A = B$. Equality of the last two would give $A \leq B$ by Lemma 3.2(i), contrary to the antichain assumption. Thus $m \geq 3 > 2$.

For $r \geq 3$, choose r covers with pairwise distinct labels. Their $\binom{r}{2}$ pairwise joins are distinct, as before. Let w be the join of all the selected covers. It cannot equal any of those pairwise joins: a third cover would lie below that pairwise join, and Lemma 3.1 would give a comparability between different labels. Also $w \in \text{Jr}(L)$. Indeed, if w were join-irreducible, every selected cover, being strictly smaller than w , would lie below its unique lower cover; their join could not be w . Hence $m \geq \binom{r}{2} + 1 > r$. $\square$

Lemma 4.2 (Congruences of a quotient). *For $\alpha \in \text{Con}(L)$, congruences of L/α correspond bijectively to congruences of L containing α .*

Proof. Pull an equivalence relation on the quotient back along the quotient map. Conversely, a congruence containing α induces a well-defined relation on α -classes. The two operations are inverse, and compatibility follows because the quotient operations are induced by those of L . $\square$

Proof of the equality characterization in Theorem 1.1. Assume $|\text{Con}(L)| = 2^{n-k-1}$ and choose a fan of k upper covers of u . Every inequality in the numerical proof must be an equality. Therefore

$$m = r, \quad q = n - k - 1, \quad |\text{Con}(L)| = 2^q. \quad (4.1)$$

By Lemma 2.5, $Q(L)$ is an antichain. Lemma 4.1 rules out $r \geq 2$, so

$$r = m = 1. \quad (4.2)$$

Let v be the unique join-reducible element, and let α be the common label of the fan. Every pairwise join of its covers is v . Consequently α identifies all elements of

$$B = \{u, a_1, \dots, a_k, v\}. \quad (4.3)$$

In detail, it identifies each a_i with u , and joining two of these identifications also identifies v with u .

We next show that B is exactly the sole nonsingleton α -class. Since all subsets of $Q(L)$ occur as signatures, exactly 2^{q-1} congruences contain α . By Lemma 4.2 and the classical bound applied to L/α ,

$$2^{n-k-2} = 2^{q-1} = |\text{Con}(L/\alpha)| \leq 2^{|L/\alpha|-1}. \quad (4.4)$$

Thus $|L/\alpha| \geq n - k - 1$. On the other hand, the $k + 2$ distinct elements in (4.3) belong to a single class, so $|L/\alpha| \leq n - k - 1$. Equality follows. Merging $k + 2$ elements already accounts for the full reduction of $k + 1$ in the quotient size; no additional element can enter that class and no other class can be nonsingleton. Convexity now gives

$$[u, v] = B \cong M_k. \quad (4.5)$$

It remains to locate every element outside this interval. First, each $x \in L$ is comparable with v . Otherwise $x \vee v$ would be a join-reducible element strictly greater than v , contradicting its uniqueness. Second, if $x \leq v$ and $x \notin B$, meet the relation $u \equiv v \pmod{\alpha}$ with x . This gives $u \wedge x \equiv x \pmod{\alpha}$. The class of x is a singleton, so $u \wedge x = x$, that is, $x \leq u$. Thus every element outside B lies below u or above v .

Both $[0, u]$ and $[v, 1]$ are chains. Incomparable elements in the former would have a join-reducible join at most $u < v$, and incomparable elements in the latter would have a join-reducible join

strictly above v . Both are impossible. Consequently $L \cong C_s \dot{+} M_k \dot{+} C_t$ for some $s, t \geq 1$, and counting the two identified endpoints gives $n = s + (k + 2) + t - 2 = k + s + t$.

Conversely, M_k is simple when $k \geq 3$. Its lower-edge labels are all equal by Lemma 3.2(i), and that common label identifies the whole lattice. Every upper-edge label equals a lower-edge label by transposition. A nontrivial congruence collapses an edge by the argument in Lemma 2.4, so its only congruences are equality and universality. Therefore $|\text{Con}(M_k)| = 2$.

Congruences of a glued sum $A \dot{+} D$ are in bijection with pairs of congruences of A and D . Restrictions give one direction. In the other, retain all classes except that the two classes containing the identified point are joined into one. Equivalently, map the two summands to their respective quotient lattices and glue the quotients at the image of the common point. This map is a lattice homomorphism: within each summand it is a quotient map, and across summands meet selects the lower argument while join selects the upper. Its kernel is exactly the described partition, and its restrictions recover the original congruences. These constructions are inverse. It follows that

$$|\text{Con}(C_s \dot{+} M_k \dot{+} C_t)| = 2^{s-1} \cdot 2 \cdot 2^{t-1} = 2^{s+t-1} = 2^{n-k-1}. \quad (4.6)$$

Finally, order duality interchanges the two exterior chains and preserves M_k , so the equality family is the same for lower covers. $\square$

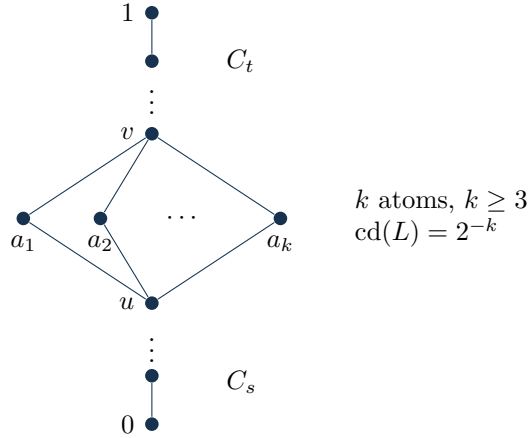


Figure 1: All equality cases. Either exterior chain may consist of its glued endpoint alone. The omitted intermediate atoms have the same two incident edges as the displayed atoms.

5 A sharp gap below the maximum

Lemma 5.1 (A missing ideal costs at least one quarter). *If a q -element poset P is not an antichain, then $|\text{Idl}(P)| \leq 3 \cdot 2^{q-2}$.*

Proof. Choose distinct $a < b$ in P . No ideal can contain b while omitting a . Exactly one of the four inclusion patterns on $\{a, b\}$ is therefore forbidden. The remaining $q - 2$ elements have at most 2^{q-2} patterns. $\square$

Theorem 5.2 (The second extremal value). *Let $k \geq 3$. If an n -element lattice L has an element with at least k upper covers and is not of the form (1.4), then $n \geq k + 3$ and*

$$|\text{Con}(L)| \leq 3 \cdot 2^{n-k-3}, \quad \text{cd}(L) \leq 3 \cdot 2^{-k-2}. \quad (5.1)$$

For every $n \geq k + 3$, equality in this bound is attained.

Proof. Put $N = n - k - 1$. Section 3 gives $q \leq N$. If $q \leq N - 1$, then $|\text{Con}(L)| \leq 2^{N-1} \leq 3 \cdot 2^{N-2}$. If $q = N$, nonextremality and Lemma 2.5 imply that $Q(L)$ is not an antichain. Lemma 5.1 gives the same bound. The case $N = 1$ admits no nonextremal lattice: every nonsingleton lattice has at least two congruences, which is already the maximum in that case.

For sharpness, form T_k by adding one new element z to M_k on the edge $a_1 \prec v$, replacing it by $a_1 \prec z \prec v$. The k lower edges remain covers and all their labels are the universal congruence, as all pairwise joins are still v . The join-irreducibles are $a_1, \dots, a_k, z$. The remaining label $\beta = \text{con}(a_1, z)$ has exactly one nonsingleton block, $\{a_1, z\}$. To verify compatibility directly, a different atom has meet u with both a_1, z and join v with both; operations with u, v, a_1, z also preserve this two-element block. Thus $Q(T_k)$ is a two-element chain $\beta < \mathbf{1}_L$, and T_k has exactly three congruences. It has $k + 3$ elements. Gluing chains at the ends multiplies the number of congruences by the corresponding powers of two, leaving density unchanged. This gives every larger n . $\square$

Corollary 5.3 (Exact top two values in the constrained class). *For $n \geq k + 3$ and $k \geq 3$, among all n -element lattices having an element with at least k upper covers, the largest and second largest distinct numbers of congruences are*

$$2^{n-k-1} \quad \text{and} \quad 3 \cdot 2^{n-k-3},$$

respectively. The corresponding densities are 2^{-k} and $3 \cdot 2^{-k-2}$.

The gap is a consequence of the classical ideal representation together with the new exponent bound. It is not an assertion about the two largest congruence counts among *all* n -element lattices, and it does not classify all lattices attaining the second value.

6 More information from the fan labels

6.1 An ideal-count refinement

Let P be the poset of the r distinct fan labels α_i , with order given by inclusion of congruences. Write $I(P) = |\text{Idl}(P)|$. The following retains order information that the power-of-two bound discards.

Proposition 6.1 (Fan-profile bound). *For a selected fan of k upper covers,*

$$|\text{Con}(L)| \leq I(P) 2^{|\text{Ji}(L)|-k}, \quad \text{cd}(L) \leq I(P) 2^{-m-k}. \quad (6.1)$$

This formula is valid also for $k = 2$; the deduction $m \geq r$ is the part that requires $k \geq 3$.

Proof. Choose the k distinct representatives j_i from Lemma 2.3. Their collapse pattern is specified by an ideal of P , with repeated labels required to have the same truth value. The remaining $|\text{Ji}(L)| - k$ representatives supply at most that many independent binary choices. Together these data determine the full signature, so they determine the congruence. This is an injection into a set of size $I(P) 2^{|\text{Ji}(L)|-k}$; surjectivity is not asserted. $\square$

Since $I(P) \leq 2^r$, Proposition 6.1 recovers Corollary 3.5. It can be sharper when the fan labels have comparabilities. It can still be strict when relations to labels outside the fan forbid additional choices.

6.2 Small exact examples

The tables below are computed independently by the programs in the archive. All mathematical descriptions in this subsection are also directly checkable from the displayed orders and congruence partitions. The small lattices serve as illustrations; their construction is not claimed to be new.

Define E_6 on $\{0, 1, 2, 3, 4, 5\}$ by the cover edges

$$(0, 1), (0, 2), (0, 3), (2, 4), (3, 4), (1, 5), (4, 5).$$

Its bottom fan is $\{1, 2, 3\}$. It has two distinct labels, with $\alpha_2 = \alpha_3 < \alpha_1$. Its congruences are equality, universality, and

$$\{0, 2, 3, 4\} \mid \{1, 5\}.$$

Thus $m = r = 2$ does not suffice for equality in the main theorem: the label comparability reduces the count from four possible subsets to three ideals.

Define E_7 on $\{0, 1, 2, 3, 4, 5, 6\}$ by

$$(0, 1), (0, 2), (0, 3), (1, 4), (3, 4), (2, 5), (3, 5), (4, 6), (5, 6).$$

The three bottom-fan labels are distinct, with $\alpha_3 < \alpha_1$ and $\alpha_3 < \alpha_2$, while α_1, α_2 are incomparable. Their poset has five ideals. The three nontrivial proper congruences have blocks

$$\{0, 3\} \mid \{1, 4\} \mid \{2, 5\} \mid \{6\},$$

$$\{0, 1, 3, 4\} \mid \{2, 5, 6\},$$

$$\{0, 2, 3, 5\} \mid \{1, 4, 6\}.$$

Together with equality and universality, these give five congruences.

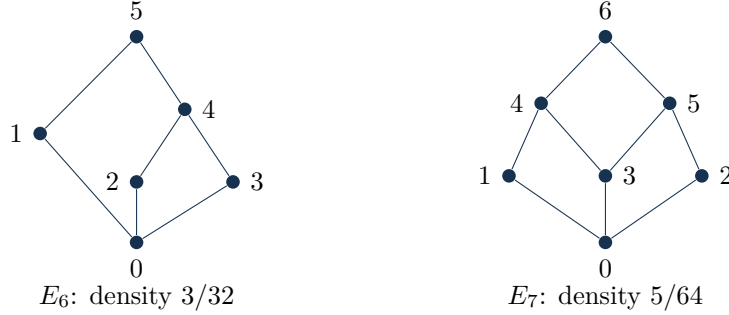


Figure 2: Two illustrative nonextremal fans. On the left two fan labels coincide and are smaller than the third. On the right all three labels are distinct, but one is below both others.

Lattice	n	k	r	m	$I(P)$	$ \text{Con} $	Density	Profile bound
M_2	4	2	2	1	4	4	1/2	1/2
M_3	5	3	1	1	2	2	1/8	1/8
M_4	6	4	1	1	2	2	1/16	1/16
E_6	6	3	2	2	3	3	3/32	3/32
E_7	7	3	3	3	5	5	5/64	5/64
T_3	6	3	1	1	2	3	3/32	1/8
B_3	8	3	3	4	8	8	1/16	1/16
$C_2 \dot{+} M_3 \dot{+} C_2$	7	3	1	1	2	8	1/8	1/8

Table 1: Exact profiles at the designated bottom fan, or at the bottom of the M_3 summand in the last row. Here B_3 means the Boolean lattice of all subsets of a three-element set, so it has eight elements. The profile bound is the right-hand side of (6.1) for density.

For the Boolean lattice B_d of all subsets of a d -element set, $\text{Ji}(B_d)$ consists of the singleton subsets. Its d labels are independent: for any collection of coordinates to forget, equality on all the other coordinates is a congruence. Hence it has exactly 2^d congruences and

$$\text{cd}(B_d) = 2^{d-2^d+1}.$$

Its bottom has d covers. This illustrates how much stronger the actual density decay can be when many different labels create many join-reducible elements.

6.3 Why two covers are exceptional

For $k = 1$, there is no improvement over density 1; chains attain it. For $k = 2$, the join of two distinct upper covers is join-reducible, so $\text{cd}(L) \leq 1/2$ by (2.6). The four-element lattice M_2 attains $1/2$, whereas $2^{-2} = 1/4$. Thus the proposed formula is genuinely false for two covers. In the proof of Lemma 3.3, the missing ingredient is exactly the third cover needed when $r = 2$.

The sharp maximum densities under an “at least k covers” constraint are therefore 1 for $k = 1$, $1/2$ for $k = 2$, and 2^{-k} for $k \geq 3$, whenever the specified size admits such a lattice. The minimum sizes are respectively 2, 4, and $k + 2$.

7 A quadratic bound for the skeleton

We now use the established cover theorem in the structural framework of the source. Following Czédli [1, Section 5, equation (5.1)], put

$$\begin{aligned} \text{Skel}(L) &= \text{Jr}(L) \cup \text{Mr}(L) \\ &\cup \bigcup_{x \in \text{Jr}(L)} \text{LoCov}_L(x) \cup \bigcup_{x \in \text{Mr}(L)} \text{UpCov}_L(x). \end{aligned} \quad (7.1)$$

Here $\text{LoCov}_L(x) = \{y : y \prec x\}$ and $\text{UpCov}_L(x) = \{y : x \prec y\}$. When nonempty, this set is a sublattice: joins and meets of comparable members are already members, while the join of incomparable members is join-reducible and their meet is meet-reducible. A chain has empty skeleton.

Theorem 7.1 (Quadratic skeleton bound). *Let $0 < p \leq 1$, put*

$$t = \lfloor \log_2(1/p) \rfloor, \quad D = \max\{2, t\}, \quad (7.2)$$

and let L be a finite lattice with $\text{cd}(L) \geq p$. Then

$$|\text{Skel}(L)| \leq 2t(D + 1) = 2t(\max\{2, t\} + 1). \quad (7.3)$$

In fact, each of $|\text{Jr}(L)|$ and $|\text{Mr}(L)|$ is at most t , and every element has at most D upper covers and at most D lower covers.

Proof. By (2.6), $\text{cd}(L) \leq 2^{-|\text{Jr}(L)|}$. Together with $\text{cd}(L) \geq p$, this yields $|\text{Jr}(L)| \leq t$. Apply the same argument to the order dual for $\text{Mr}(L)$.

Suppose an element has $d \geq 3$ upper covers. Theorem 1.1 gives $p \leq \text{cd}(L) \leq 2^{-d}$, hence $d \leq t$. Counts of zero, one, or two covers are already at most D . Duality gives the lower-cover bound. Finally, count the union in (7.1), allowing overlaps:

$$\begin{aligned} |\text{Skel}(L)| &\leq \sum_{x \in \text{Jr}(L)} (1 + |\text{LoCov}_L(x)|) + \sum_{x \in \text{Mr}(L)} (1 + |\text{UpCov}_L(x)|) \\ &\leq (|\text{Jr}(L)| + |\text{Mr}(L)|)(D + 1) \\ &\leq 2t(D + 1). \end{aligned}$$

□

Czédli’s earlier numerical estimate uses

$$k(p) = \lfloor \log_2(1/p) \rfloor + 3, \quad f(p) = 2k(p)(R_{k(p)-1}(k(p)) + 1)$$

and gives $|\text{Skel}(L)| \leq f(p)$ for $\text{cd}(L) \geq p$ [1, Lemma 5.5]. Theorem 7.1 replaces this Ramsey-dependent quantity by $O((\log(1/p))^2)$. It is a bound for exactly the same skeleton, not for a newly defined smaller invariant.

	t	0	1	2	3	4	10
	D	2	2	2	3	4	10
Skeleton upper bound		0	6	12	24	40	220

Table 2: Values of the proved skeleton bound. The $t = 0$ case says that density strictly greater than $1/2$ forces a chain. The table gives upper bounds, not claims that each is optimal.

The theorem does not bound $|L|$ in terms of p . Arbitrarily long chains have density 1, and adding exterior chains preserves any lattice’s density by the glued-sum formula. The source’s structural lemmas explain how skeletons encode branching while chain segments can remain unbounded [1, Lemmas 5.2–5.4]. The quadratic estimate concerns that bounded branching part only.

8 Exact computational verification

8.1 Enumerating every small lattice without a database

The verifier uses Python’s standard library and integer arithmetic. A *natural labelling* of a poset on $\{0, \dots, h-1\}$ means that $x <_P y$ implies $x < y$ as integers. Such posets can be generated recursively: after constructing the first h vertices, the strict lower set of the new, numerically largest vertex can be any order ideal of the existing poset. Every natural poset is generated exactly once, because that lower set and the induced old poset uniquely determine its final extension.

For $n \geq 2$, the program enumerates natural posets on $n-2$ interior elements and adds a bottom labelled 0 and a top labelled $n-1$. It retains precisely the orders in which every pair has a greatest lower bound and a least upper bound. The singleton is treated separately. Every finite lattice admits a natural labelling with these bottom and top labels, by choosing a linear extension of its order. Therefore this process includes every isomorphism class. It does *not* eliminate isomorphic natural labellings, and its counts must not be read as counts of isomorphism classes.

Downsets and upsets are integer bitmasks. Meet and join tables, cover sets, join-irreducibles, and the skeleton are computed from the order, not supplied by a lattice database or inferred from a proposed family description.

8.2 Two congruence computations

The main computation obtains $\text{con}(a, b)$ by starting with the equality relation, merging a, b , and repeatedly closing under meet and join with every lattice element. A union-find data structure maintains equivalence closure. The process terminates because each effective merge reduces the number of blocks. Every merge is forced in any congruence identifying a, b , and the final partition is compatible; hence it is exactly the principal congruence.

All congruences are then generated as joins of the distinct labels γ_j . The join is the equivalence closure of the union of two partitions. Lemma 2.4 proves completeness of this generation method. The program also directly checks the compatibility of every generated partition.

For every lattice of at most seven elements, a second computation enumerates *all set partitions* and tests compatibility directly, without using principal labels or the generating theorem. The resulting sets of partitions are compared for equality, not merely for equal cardinality. This independently checks the congruence enumeration on 371 naturally labelled lattices. Both methods are ordinary executable mathematics, not formal proofs checked by a proof-assistant kernel.

8.3 What was checked and what completed

For each retained lattice, the program checks every selected subset of at least three upper covers of every vertex. In particular, it checks fans that are proper subsets of larger cover sets. For every such selection it verifies the collision identities, $m \geq r$, the repeated-label exponent, the main bound, the profile bound, the equality classification by direct order recognition, and the nonextremal gap. It also checks the dual numerical bound and the skeleton bound at the exact rational value $p = \text{cd}(L)$.

n	Interior posets	Lattices	Congruences	Fans	Equality fans	Second method
1	1	1	1	0	0	1
2	1	1	2	0	0	1
3	1	1	4	0	0	1
4	2	2	12	0	0	2
5	7	7	49	1	1	7
6	40	39	281	17	3	39
7	357	320	2188	295	6	320
8	4824	3637	22743	5979	10	0
Total		4008	25280	6292	20	371

Table 3: Completed exhaustive verification. “Congruences” is the sum of congruence counts over the naturally labelled lattices in that row. “Second method” counts lattices checked by all-partition enumeration. For $n = 1$, the interior-poset column uses the program’s empty-interior convention.

All checks in Table 3 passed. Of the 6,292 fans, 1,137 have exactly two distinct labels and 108 have at least three; thus the tests include the mixed-label cases essential to the argument, not merely simple diamonds. The maximum densities for fan sizes 3, 4, 5, 6 in the completed range are $1/8, 1/16, 1/32, 1/64$, respectively, as predicted.

Separately, the example program checks 35 structured cases: M_k for $3 \leq k \leq 20$; four diamond-with-chain examples; the one-edge subdivisions T_k for $3 \leq k \leq 12$; and the Boolean lattices B_3, B_4, B_5 . The last has 32 elements. These are selected larger examples, not an exhaustive 32-element enumeration. Their full-fan profiles and exact counts are in `data/structured_checks.json`.

The finite tests corroborate the arguments but do not replace them. The all- n statements depend on the written proofs. An exploratory run at $n = 9$ did not finish within the allotted tool time and is not included in the completed scope or totals.

8.4 Reproducing the archive

From the archive’s root directory, run

```
python3 code/verify.py --max-n 8 --independent-through 7
python3 code/examples.py
latexmk -pdf -interaction=nonstopmode -halt-on-error sharp_cover_bound.tex
```

Python 3.10 or later is required for the type-annotation syntax; the recorded run used Python 3.13. No numerical library, network connection, lattice database, or external theorem prover is needed for verification. The L^AT_EX document uses standard AMS, Latin Modern, TikZ, and layout packages. The recorded elapsed times are machine-dependent diagnostics; the mathematical outputs use exact integers, partitions, and rational numbers.

The random-selection script is deliberately not part of the verification command. Its recorded draw is an input to the project, not a theorem to be regenerated. Re-running a genuinely random

draw would not reproduce the same outcome. The script refuses to overwrite the existing selection record.

9 Proof audit and remaining scope

The main logical dependencies can be summarized without a computational step:

$$\text{transposition} \implies \text{join collisions} \implies m \geq r,$$

$$\text{edge representatives} \implies q \leq n - 1 - m - k + r, \quad \text{signatures} \implies |\text{Con}(L)| \leq 2^q.$$

Combining the last two inequalities with $m \geq r$ proves the target. The equality proof additionally uses antichain signatures and the elementary quotient correspondence. The sharp gap uses the classical ideal representation. The skeleton bound then follows by counting reducible elements and their covers.

Several possible pitfalls are avoided explicitly. The transposed upper interval need not be prime. The fan representatives really are distinct even when their labels coincide. A collision of disjoint pairs is handled separately from an incident collision. The two-label case needs a repeated label and therefore a third cover. The equality proof rules out additional elements inside the diamond using quotient cardinality, not an unjustified assumption that a sublattice is an interval. Finally, a skeleton bound is not a size bound for the whole lattice.

The proposed solution settles the stated all- k implication within the argument presented here. Independent specialist review would still be valuable, particularly for the equality classification and priority assessment. No claim is made of a Lean formalization, a new congruence-lattice representation theorem, a full density-spectrum classification, or an optimal skeleton bound.

Two natural continuations remain. One is to improve the quadratic skeleton estimate or to construct examples showing its correct asymptotic order. The other is to characterize every lattice attaining the second constrained value $3 \cdot 2^{n-k-3}$. The subdivision family proves that this value is sharp, but the present argument does not assert that it exhausts the second-level extremizers. A further multi-fan estimate would have to track shared labels; one cannot simply multiply separate fan bounds without controlling their dependence.

A The random area selection

The program used `secrets.randbelow(80)`, which obtains randomness from the operating system. It returned the zero-based index 9, hence the one-based index 10, *lattice theory*. The ordered list below was fixed in the script before the draw. Only one draw was made. The JSON record contains the list, method, date, indices, selected area, and a zero redraw count. It is a transparent execution record, not an independently certified random experiment or a cryptographic commitment.

- | | |
|----------------------------------|-------------------------------|
| 1. Finite group theory | 41. Graph coloring |
| 2. Infinite group theory | 42. Graph domination |
| 3. Semigroup theory | 43. Graph enumeration |
| 4. Commutative algebra | 44. Hypergraph theory |
| 5. Noncommutative algebra | 45. Ramsey theory |
| 6. Ring theory | 46. Extremal set theory |
| 7. Representation theory | 47. Permutation patterns |
| 8. Lie algebras | 48. Enumerative combinatorics |
| 9. Universal algebra | 49. Algebraic combinatorics |
| 10. Lattice theory | 50. Polyhedral combinatorics |
| 11. Order theory | 51. Discrete geometry |
| 12. Category theory | 52. Convex geometry |
| 13. Type theory | 53. Combinatorial topology |
| 14. Proof theory | 54. Knot theory |
| 15. Computability theory | 55. Low-dimensional topology |
| 16. Model theory | 56. General topology |
| 17. Set theory | 57. Metric geometry |
| 18. Boolean algebras | 58. Fractal geometry |
| 19. Automata theory | 59. Real analysis |
| 20. Formal languages | 60. Complex analysis |
| 21. Combinatorics on words | 61. Geometric function theory |
| 22. Symbolic dynamics | 62. Harmonic analysis |
| 23. Topological dynamics | 63. Functional analysis |
| 24. Ergodic theory | 64. Operator theory |
| 25. Discrete dynamical systems | 65. Matrix analysis |
| 26. Arithmetic dynamics | 66. Special functions |
| 27. Elementary number theory | 67. Orthogonal polynomials |
| 28. Additive number theory | 68. Approximation theory |
| 29. Multiplicative number theory | 69. Asymptotic analysis |
| 30. Diophantine approximation | 70. Analytic combinatorics |
| 31. Diophantine equations | 71. Generating functions |
| 32. p -adic analysis | 72. Integer sequences |
| 33. Finite fields | 73. Recurrence relations |
| 34. Coding theory | 74. Probability inequalities |
| 35. Design theory | 75. Discrete probability |
| 36. Finite geometry | 76. Random walks |
| 37. Matroid theory | 77. Stochastic processes |
| 38. Extremal graph theory | 78. Combinatorial game theory |
| 39. Algebraic graph theory | 79. Optimization |
| 40. Spectral graph theory | 80. Numerical analysis |

B Compact verification algorithms

The executable implementations are in `code/lattice_tools.py`. The following pseudocode displays the two essential closure procedures. It deliberately does not enumerate ideals of a conjectured label order and then assume that they are congruences.

```
principal_congruence(L, a, b):
    P := the equality partition of L
    merge the blocks containing a and b
    repeat until no block changes:
        for x, y in the same block of P:
            for z in L:
                merge the blocks of meet(x,z) and meet(y,z)
                merge the blocks of join(x,z) and join(y,z)
    return P

all_congruences(L):
    labels := distinct principal_congruence(L, j_*, j)
    for all join-irreducibles j
    output := {equality partition}
    for gamma in labels:
        output := output union
            {equivalence_closure(theta union gamma)
             : theta in output}
    return output
```

The independent check instead generates every set partition P and retains it if, whenever x, y share a block, the pairs $x \wedge z, y \wedge z$ and $x \vee z, y \vee z$ share blocks for every z . Testing one argument at a time suffices for compatibility in both arguments.

For equality recognition, the program does not use the number of congruences. At the designated vertex u , it checks that there are exactly k upper covers, that all their pairwise joins are one vertex v , that every other vertex lies below u or above v , and that those two exterior intervals are chains. This order-theoretic test is compared with numerical equality, so the equality classification is not tested by a circular numerical definition.

Archive map. `sharp_cover_bound.tex` and `sharp_cover_bound.pdf` are the article. `README.md` supplies commands and a result summary. `STATUS.md` separates proved claims, executed tests, and unverified status. `notes/` contains the source and proof audits. `code/` contains the selection, enumeration, and example programs. `data/` contains the original selection record, per-size verification records, aggregate totals, worked examples, and structured larger checks.

References

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- [3] Gábor Czédli, *Lattices with congruence densities larger than $3/32$* , author manuscript dated 28 July 2026; earlier version arXiv:2602.04321. <https://arxiv.org/abs/2602.04321>. Consulted author version: <https://www.math.u-szeged.hu/~czedli/m/publ.pdf/czedli-cd-3-32.pdf>.
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